EASI Desktop Metrics

Unnamed stream

Reach	NHDPlusID 10000900049512
Analysis point	43.68578, -72.23671
COMID (StreamCat reach)	9327042
Drainage area (km2)	1.0005
NARS-9 region	NAP
Slope class	0.5_to_2
NHD feature code	46003
Ecosystem Condition Index	0.67
Physical, chemical, biological	0.65, 0.63, 0.74
Functions rated	20 of 20

#	Discipline	Function	Metric	Entries (value rated)	Rating	Source
1	Hydrology	Catchment hydrology	Watershed Land-Cover Pressure	Watershed impervious cover: **6.03%** Watershed agricultural cover: **2.11%**	Good	STAF site engine v0.5.0, HR reach watershed: NLCD 2021 impervious cover + STAF site engine v0.5.0, HR reach watershed: NLCD 2021 crop + hay
2	Hydrology	Surface water storage	Percent Wetlands in Watershed	Woody wetland cover: **11.81%** Herbaceous wetland cover: **1.35%**	Good	STAF site engine v0.5.0, HR reach watershed: NLCD 2021 wetlands
3	Hydrology	Reach inflow	Concentrated Runoff / Stormwater Inputs	Watershed road density: **1.7055 km/km²**	Fair	STAF site engine v0.5.0, HR reach watershed: TIGERweb road density
4	Hydrology	Streamflow regime	Flow Alteration (Regulation / Water Use)	Upstream normalized storage: **0 m³/km²** Annual runoff: **700.9 mm**	Good	STAF site engine v0.5.0, HR reach watershed: NID normal storage + EROM-derived runoff
5	Hydraulics	Low flow and baseflow dynamics	Low-flow Wetted Connectivity†	Monthly flow variability: **0.851775**	Good	NHDPlus V2 EROM modeled monthly flows
6	Hydraulics	High flow dynamics	Floodplain Engagement Frequency (bankfull recurrence)	Bank-height ratio: **1.7**	Poor	USGS 3DEP reach cross-sections (median of 9)
7	Hydraulics	Floodplain connectivity	Floodplain Access / Entrenchment	Entrenchment ratio: **1.54**	Good	USGS 3DEP reach cross-sections (median of 9)
8	Hydraulics	Hyporheic connectivity	Hyporheic Exchange Indicators	Channel slope: **0.01 m/m** Reach sinuosity: **1.061**	Good	NHDPlus slope + selected reach geometry
9	Geomorphology	Channel evolution	Channel Evolution Stage & Trends	Bank-height ratio: **1.7** Entrenchment ratio: **1.54**	Poor	USGS 3DEP reach cross-sections (median of 9) + NHDPlus FCODE
10	Geomorphology	Channel and floodplain dynamics	Bank Erosion & Armoring Condition	Bank-height ratio: **1.7**	Poor	USGS 3DEP reach cross-sections (median of 9)

#	Discipline	Function	Metric	Entries (value rated)	Rating	Source
11	Geomorphology	Sediment continuity	Sediment Supply Potential (watershed/banks)	Agricultural cover: **2.11%** Soil erodibility K-factor: **0.307** Road density: **1.7055 km/km²**	Fair	STAF site engine v0.5.0, HR reach watershed: agriculture + K-factor + road density
12	Geomorphology	Bed composition and large wood	Substrate Condition (grain size/embeddedness/fines/consolidation)	Watershed agricultural cover: **2.11%**	Good	STAF site engine v0.5.0, HR reach watershed: NLCD 2021 agriculture
13	Physicochemistry	Light & thermal regime	Stream Temperature	Woody riparian cover: **68.8%** Watershed impervious cover: **6.03%**	Fair	STAF site engine v0.5.0, HR reach watershed: woody riparian cover + watershed impervious cover
14	Physicochemistry	Carbon processing	Detrital Processing (CPOM retention / shredders)	Riparian forest: **48.4%** Riparian shrub: **0%** Riparian grassland: **0%** Riparian wetland: **24.7%**	Fair	STAF site engine v0.5.0, HR reach watershed: NLCD 2021 riparian land cover (100 m)
15	Physicochemistry	Nutrient cycling	Nitrogen & Phosphorus Concentrations	Total nitrogen: **0.236 mg/L** Total phosphorus: **0.0091 mg/L**	Good	Water Quality Portal + EPA NARS nine-region thresholds
16	Physicochemistry	Water & soil quality	Regulatory Impairment Status (305b/303d/TMDL)	ATTAINS integrated-report category: **4A**	Fair	EPA ATTAINS
17	Biology	Habitat provision	In-stream Habitat Complexity & Cover	Woody riparian cover: **68.8%**	Fair	STAF site engine v0.5.0, HR reach watershed: NLCD 2021 woody riparian cover (100 m)
18	Biology	Population support	Biological Integrity (IBI / community condition)†	Predicted probability of Good BMMI condition: **0.793333**	Good	EPA StreamCat prg_bmmi0809 (areaOfInterest other)
19	Biology	Community dynamics	Invasive / Non-native Species Presence	Established NAS taxa count: **0 recorded taxa**	Good	USGS NAS (HUC12)
20	Biology	Watershed connectivity	Fish Passage & Barrier Effects (longitudinal connectivity)	Mapped NID dams within one mile: **0 mapped dams**	Good	USACE NID

† Marked desktop evidence comes from the nearest StreamCat reach downstream.

Note. This stream is outside the StreamCat network. Watershed metrics use the HR reach watershed (1.00 km²) from the STAF site engine.

The entries are the values the EASI calculator takes in column J of each scenario worksheet, as the application rated them. Generated from national datasets. A desktop screening estimate with per-metric confidence, not a field-validated assessment.